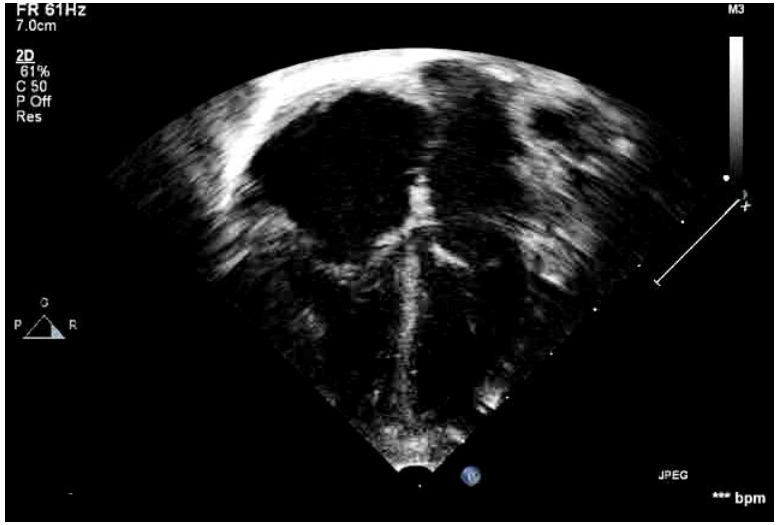
Heart Failure in Children

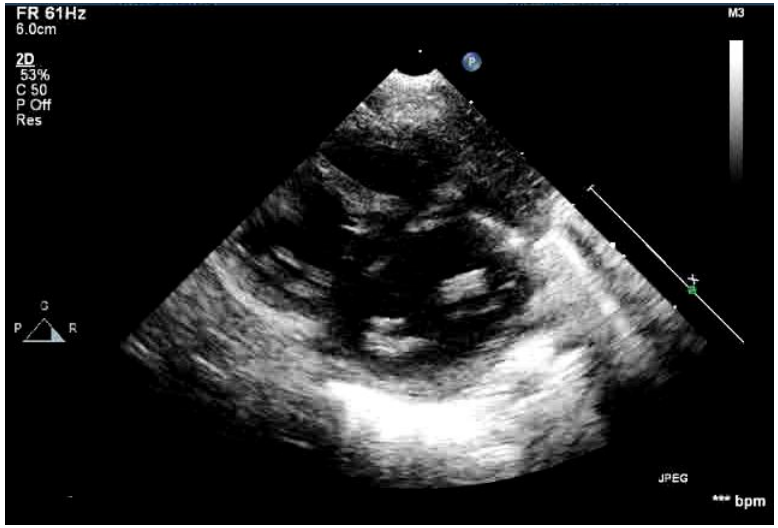
Professor Yiu-fai Cheung
MD, FRCP

Division of Paediatric Cardiology
Department of Paediatrics and Adolescent Medicine
The University of Hong Kong

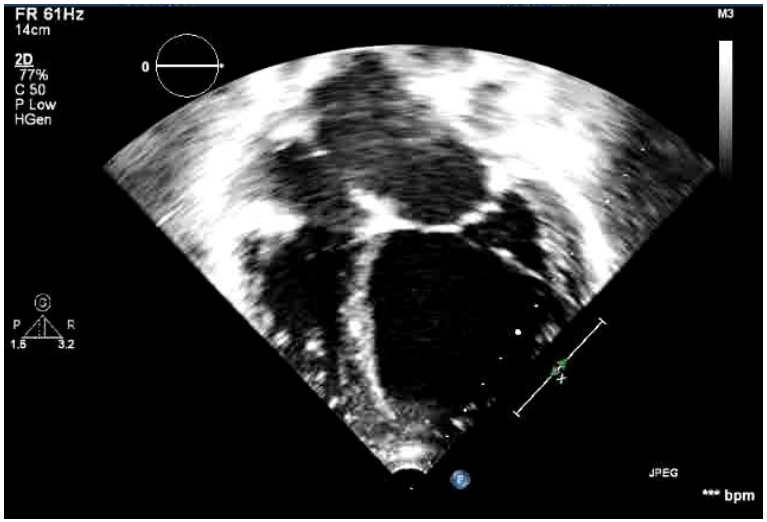
Concept of Paediatric Heart Failure

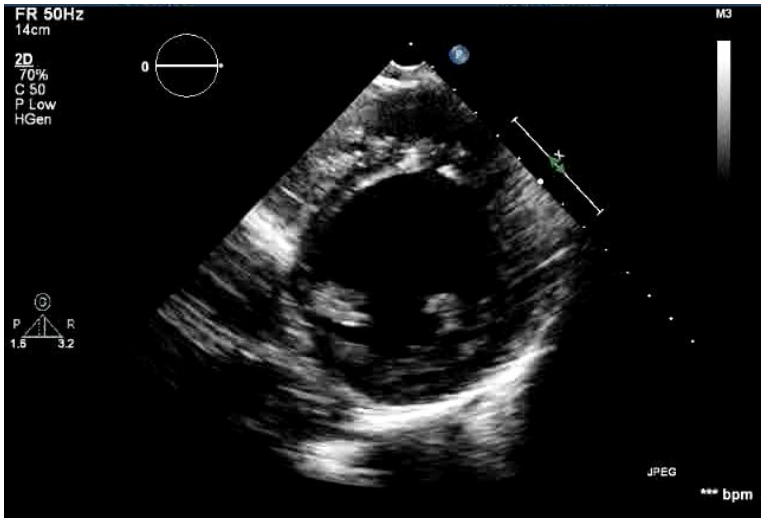
Patient A





Patient B





Concept of Heart Failure

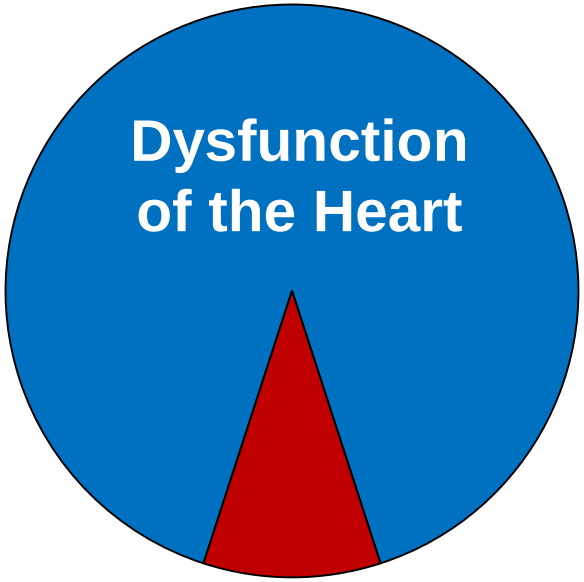
The pathophysiologic state in which the heart fails to pump blood at a rate to meet the demand of the body

Supply *vs* Demand

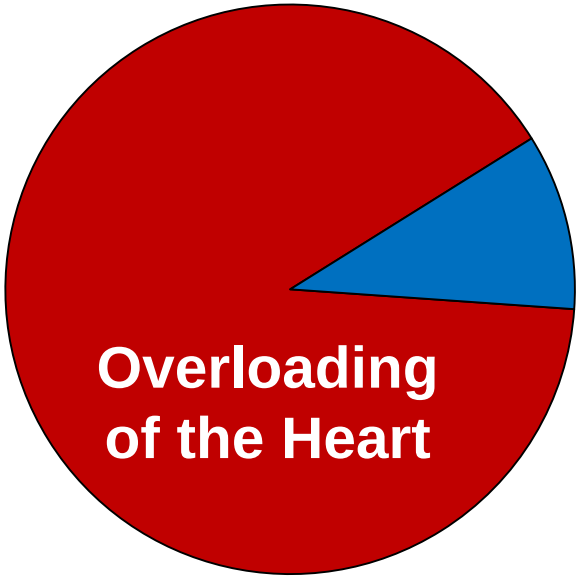
Dysfunction of the Heart
vs
Cardiac Overload
(pressure or volume overload)

‘Heart Failure Syndrome’

Adults



Children



How do Infants and Children with Heart Failure Present ?

History Taking

Symptoms

- breathlessness, especially on exertion
- poor feeding
- excessive sweating
- failure to thrive
- recurrent chest infection
- exercise incapacity in older children

Physical Examination

Signs

pulmonary venous congestion

- tachypnoea
- subcostal insucking
- wheezing in infants

systemic venous congestion

- hepatomegaly
- distension of neck veins (not obvious in small children)
- peripheral edema (rare in the absence of cardiac dysfunction)

Physical Examination

Signs

compensatory mechanisms

- tachycardia
- cardiomegaly

diminished cardiac output (unusual, only when decompensated)

- cool extremities
- prolonged capillary refill
- decreased pulse volume

Causes of Paediatric Heart Failure in Neonates, Infants and Older Children

Heart Failure in Neonates

1) Left ventricular outflow tract obstruction
2) Myocardial dysfunction
3) Abnormalities of heart rate or rhythm
4) Extra-cardiac causes

Heart Failure in Neonates

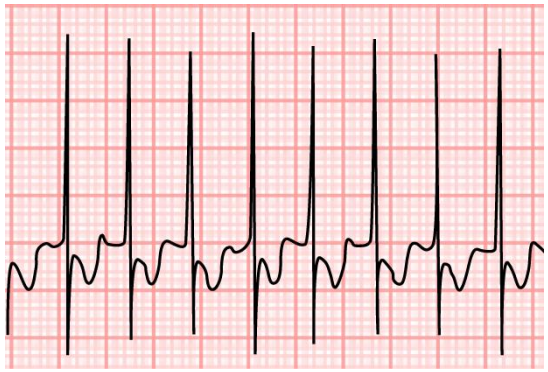
Myocardial dysfunction

- transient myocardial ischaemia
- myocarditis
- cardiomyopathy

Heart Failure in Neonates

Abnormal heart rate

supraventricular tachycardia



complete heart block

Heart Failure in Neonates

Extra-cardiac causes

- systemic disturbance causing ventricular dysfunction
  - sepsis
  - asphyxia
  - hypocalcaemia
- anaemia

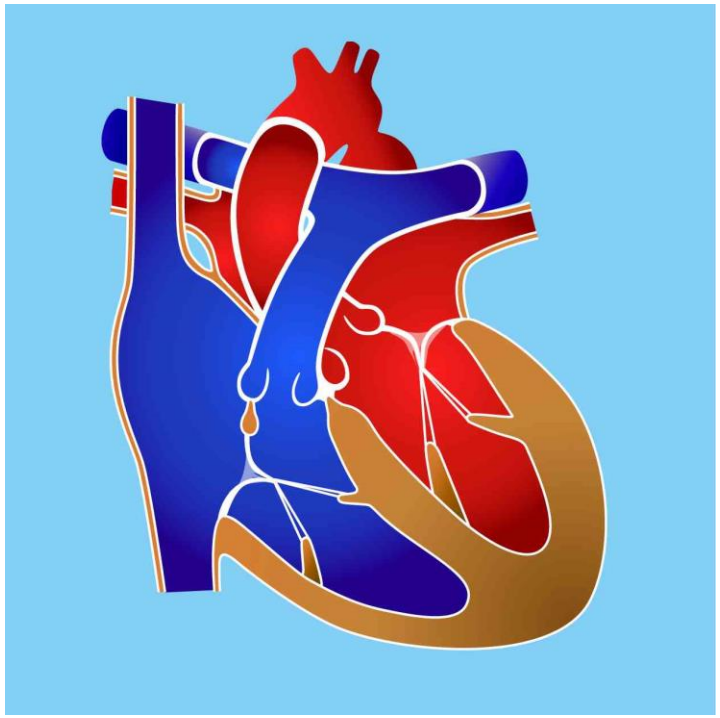
Left Ventricular Outflow Obstruction

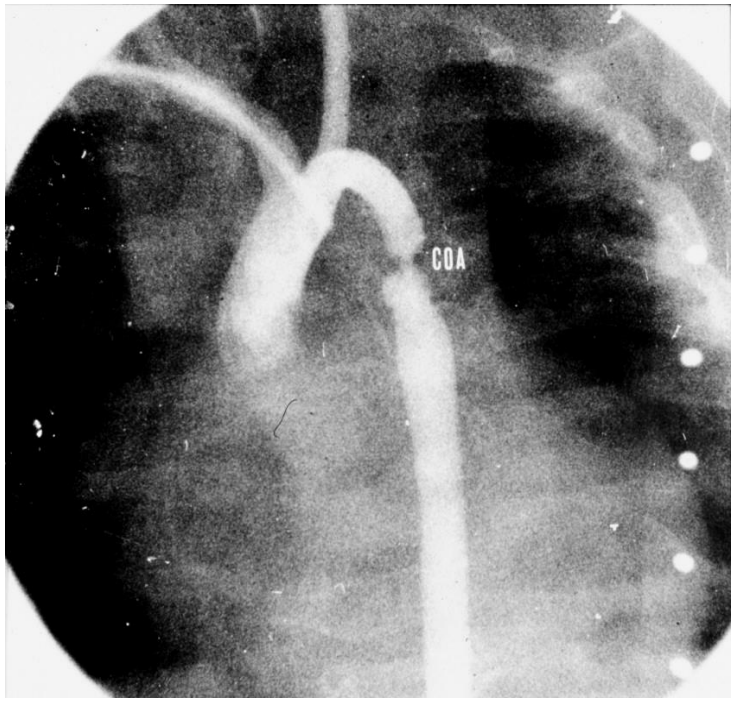
1) Coarctation or interruption of the aorta

2) Critical aortic stenosis

3) Aortic atresia (hypoplastic left heart syndrome)

Coarctation of The Aorta





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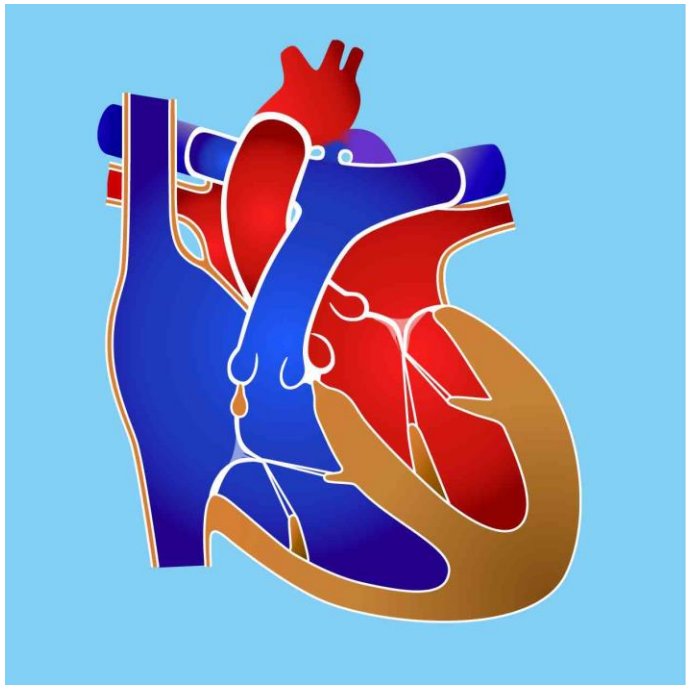
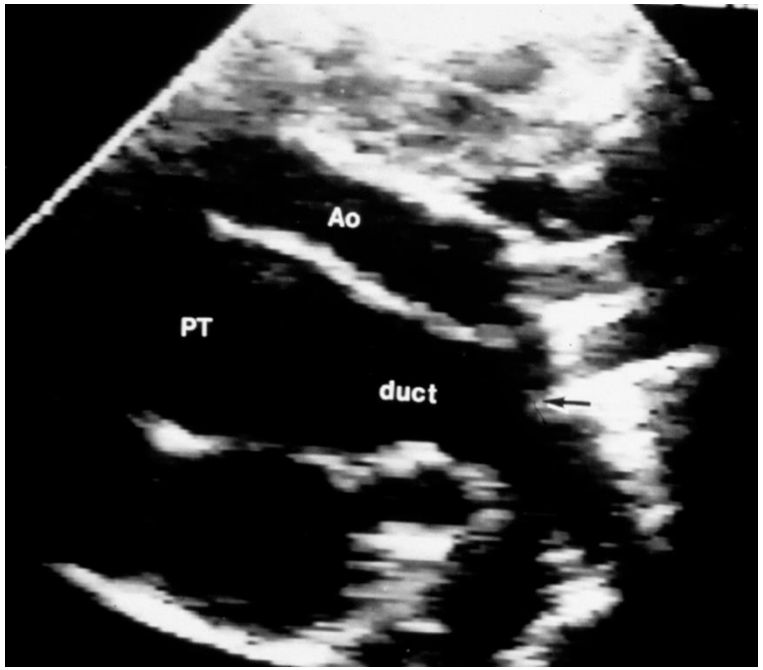
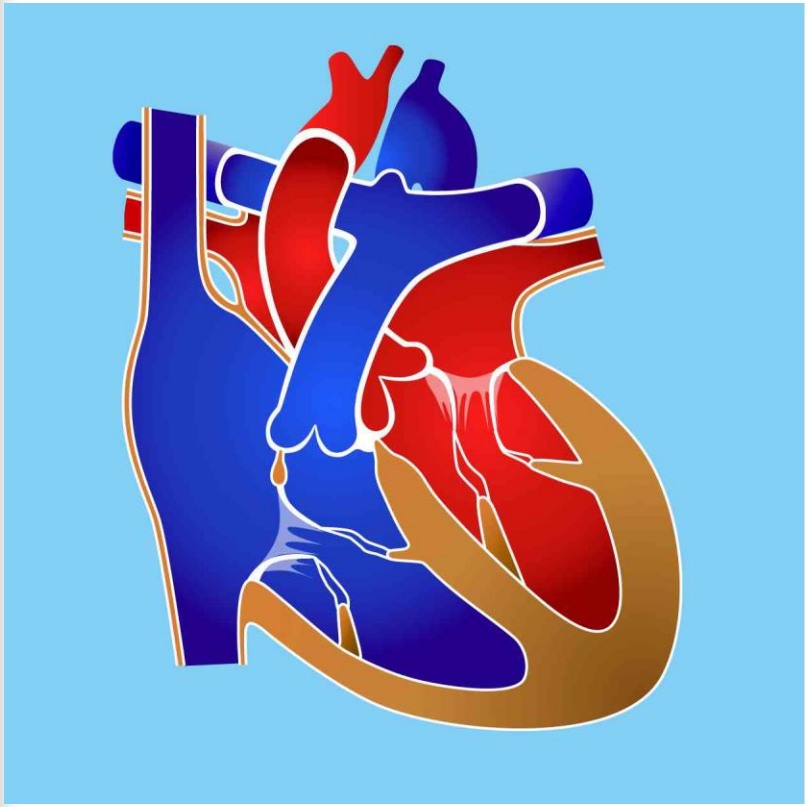
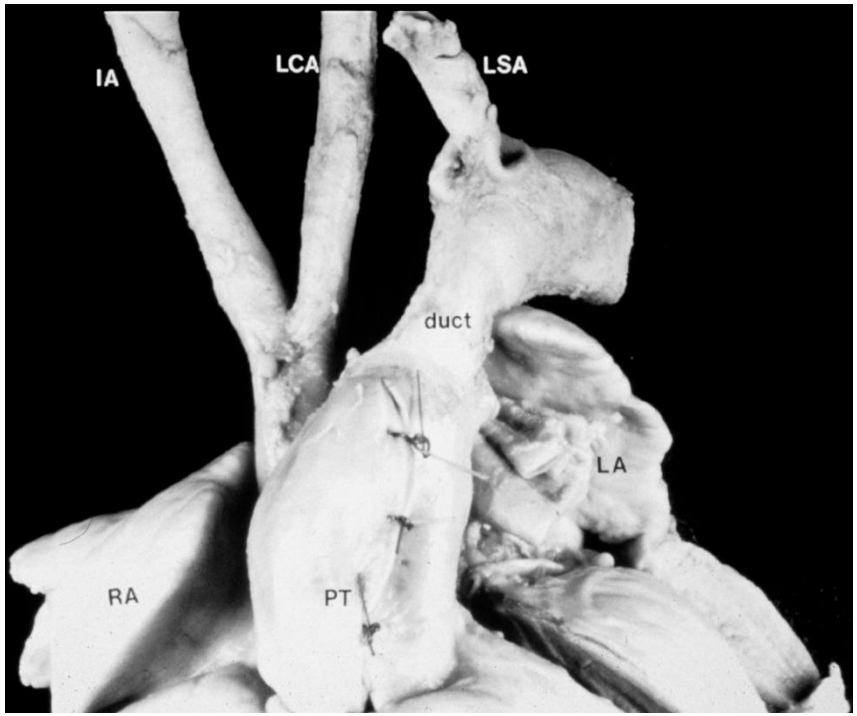




Interrupted Aortic Arch





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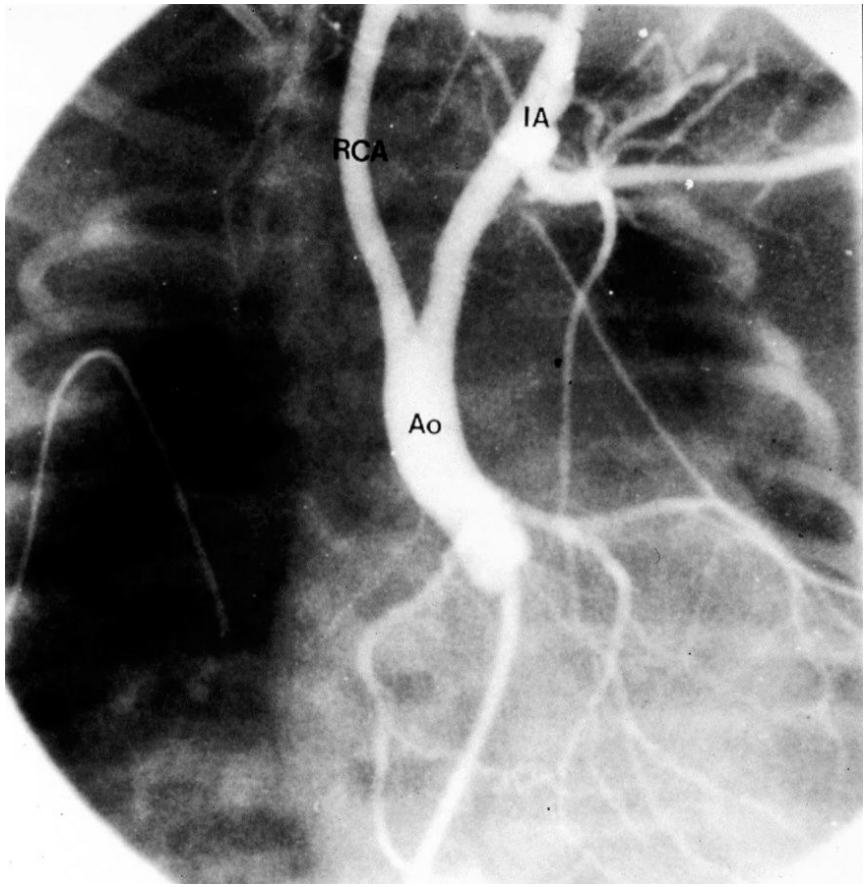
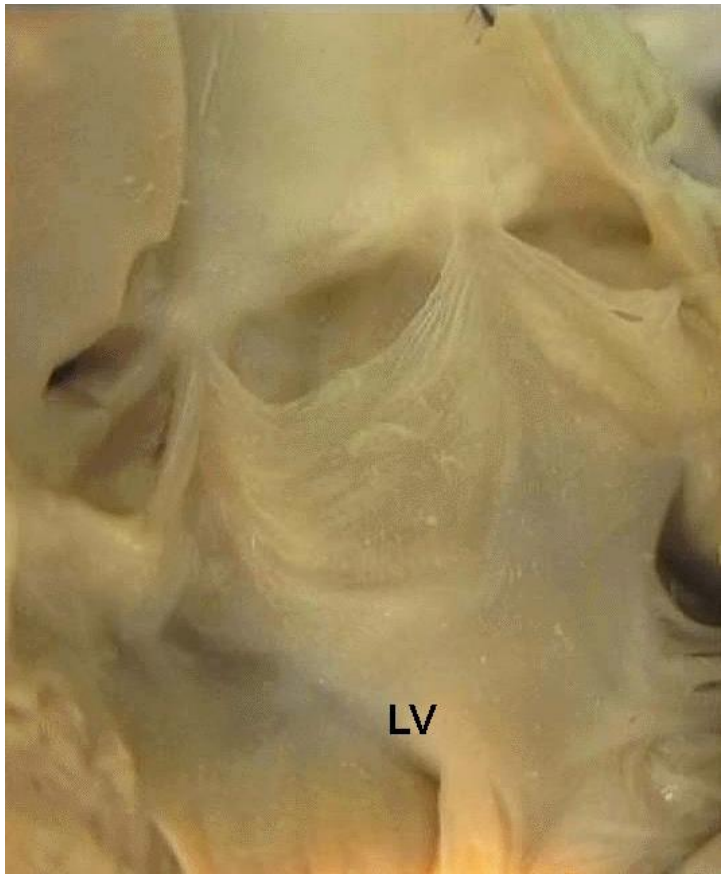
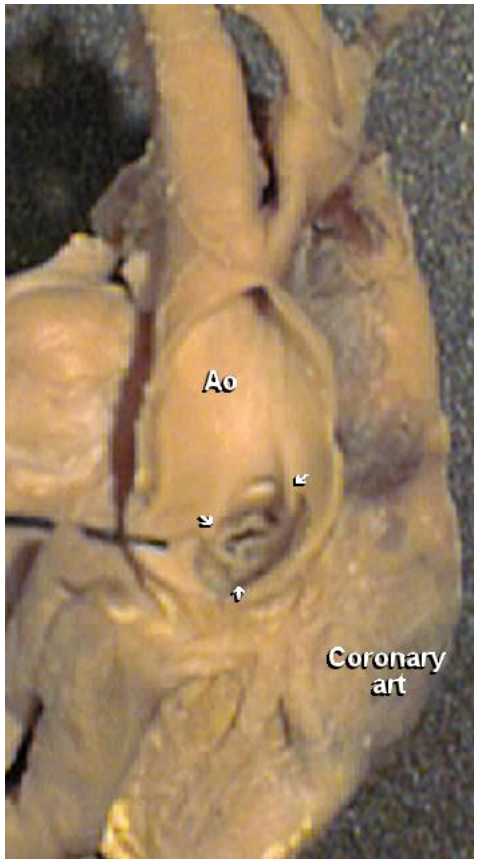


Critical Aortic Stenosis

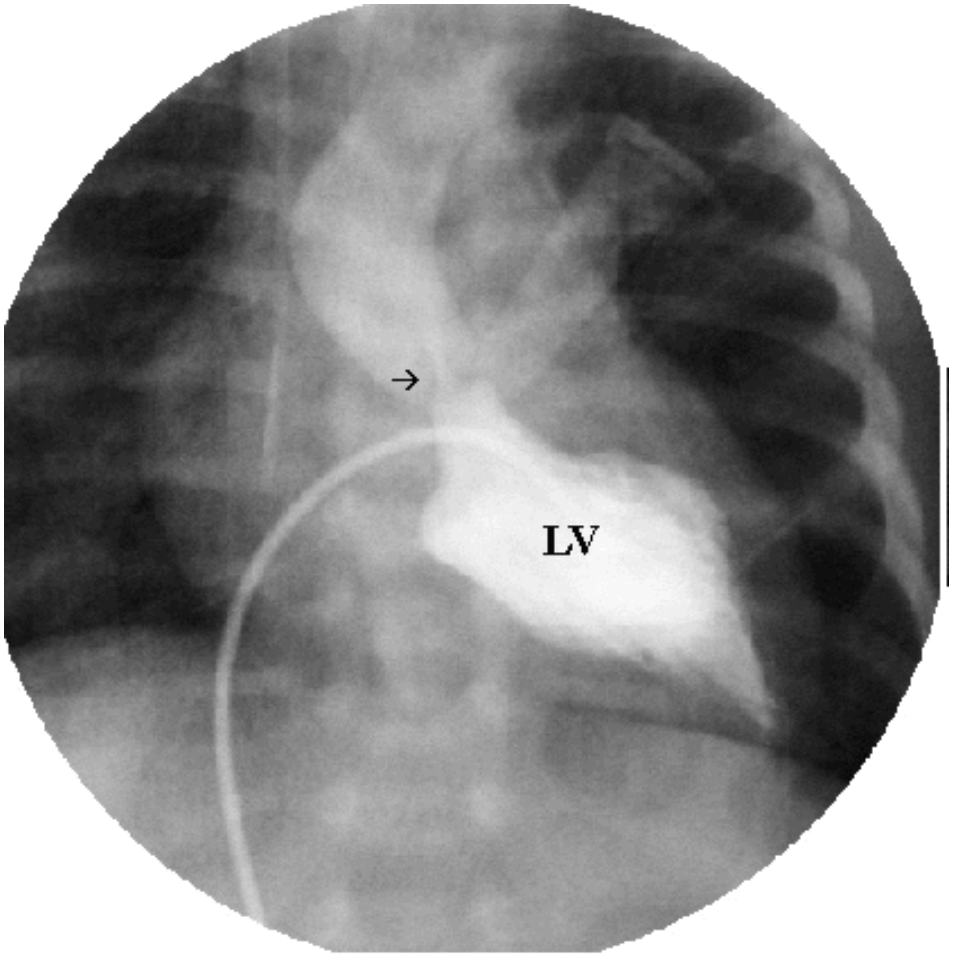


normal aortic cusps

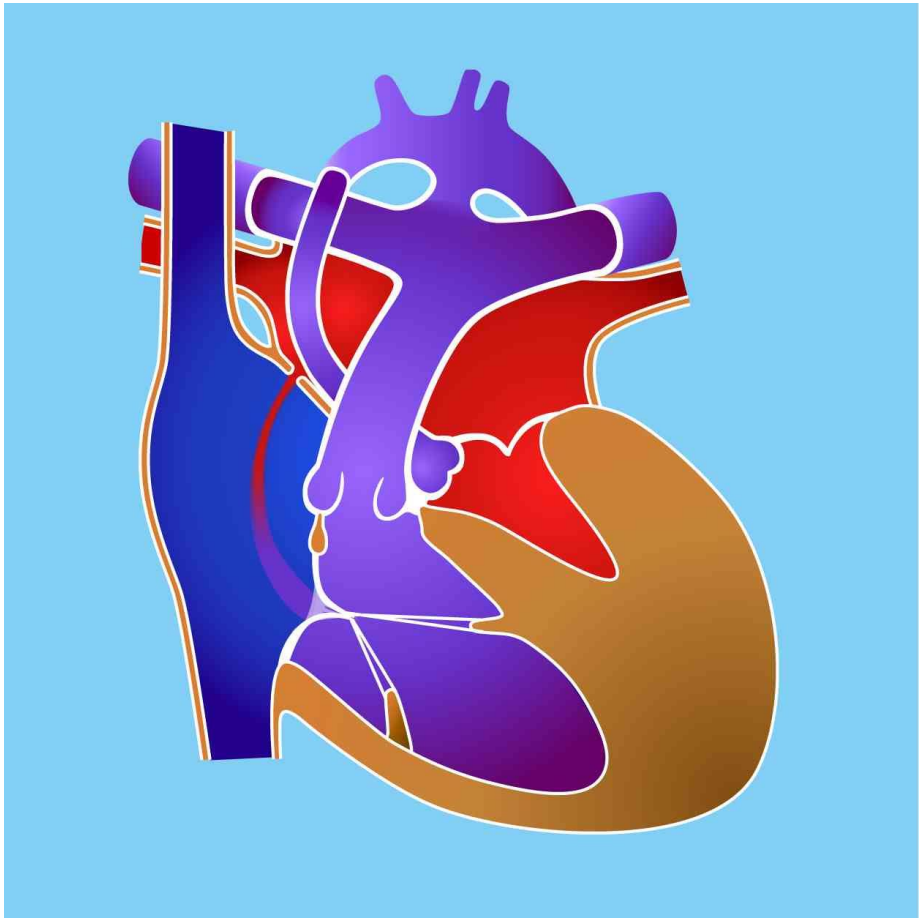


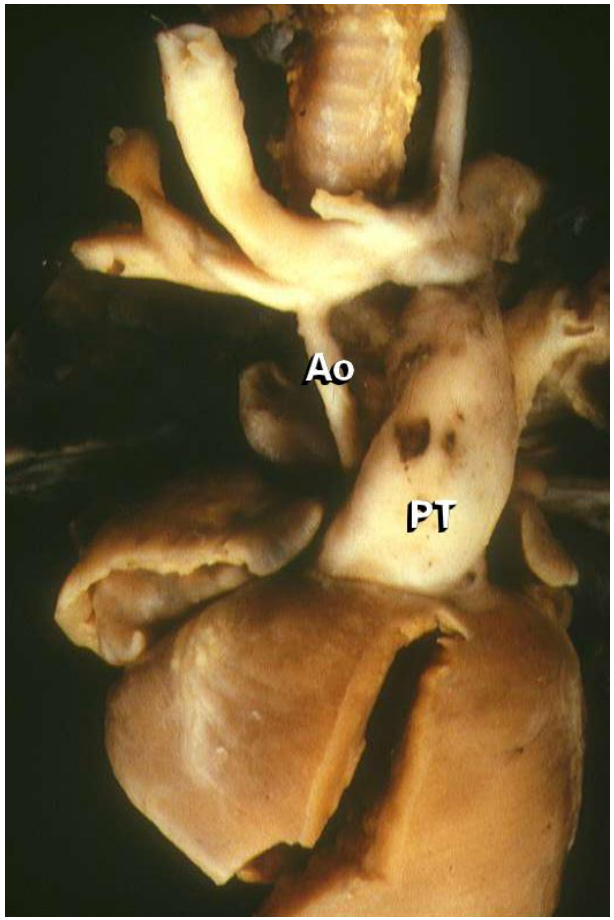
thickened and fused aortic cusps

Critical Aortic Stenosis



Hypoplastic Left Heart Syndrome





Importance of pulse examination

Femoral pulse in particular

Concept of ‘ductal-dependent systemic circulation’

Prostaglandin E1 /E2

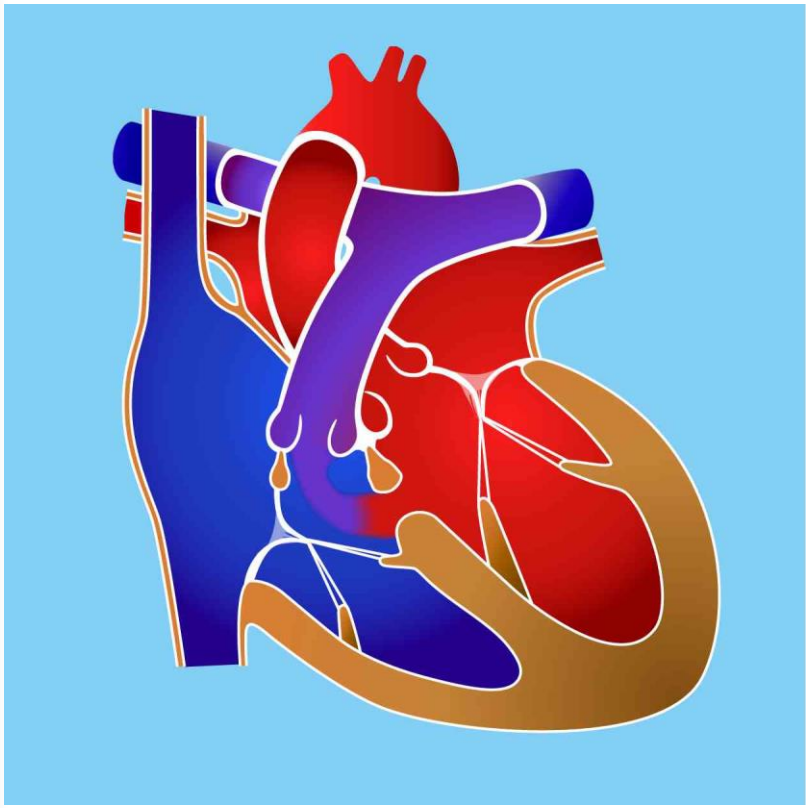
Heart Failure in Infants

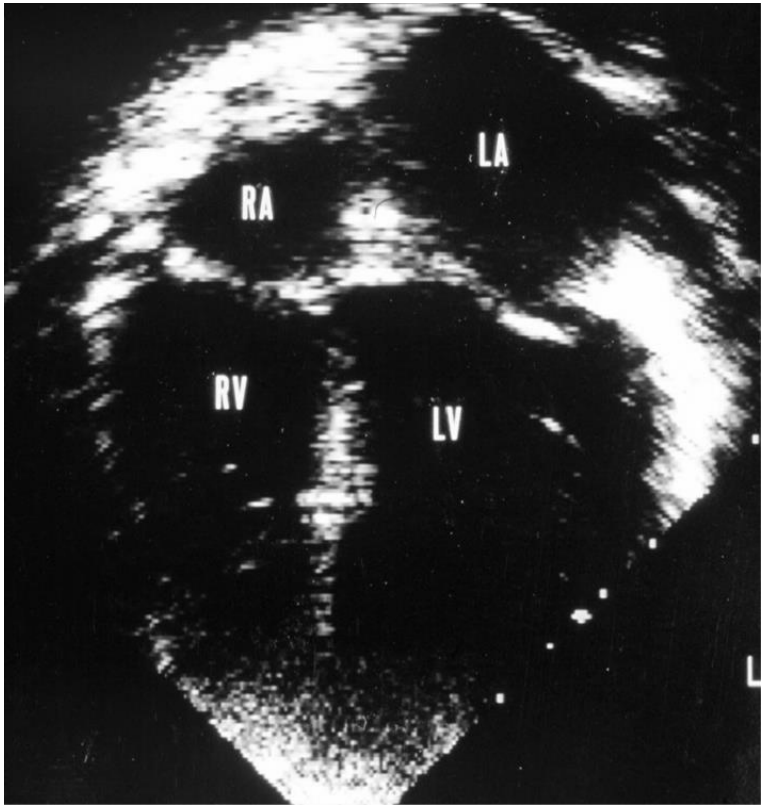
Large left-to-right shunts

- ventricular septal defect
- atrioventricular septal defect
- persistent arterial duct

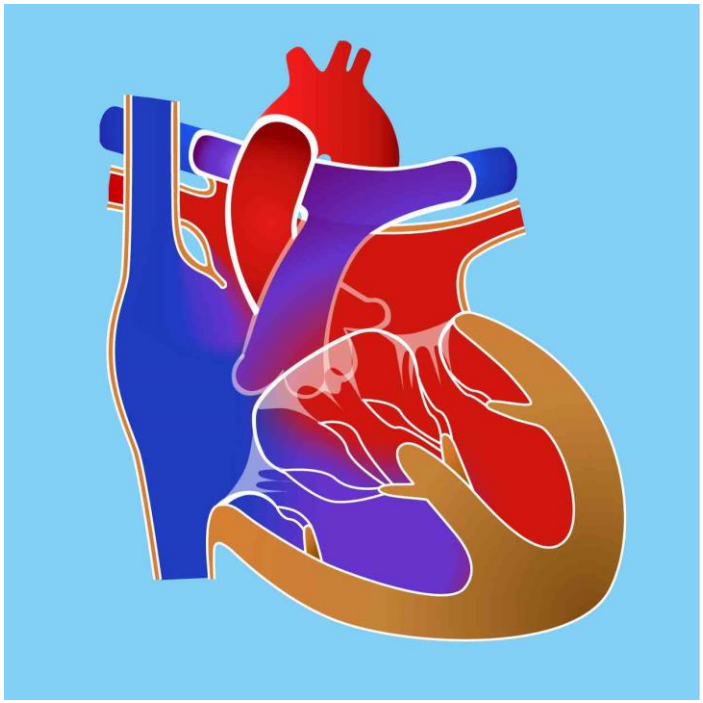
Later onset of symptoms (as compared to left ventricular outflow obstructive lesions)

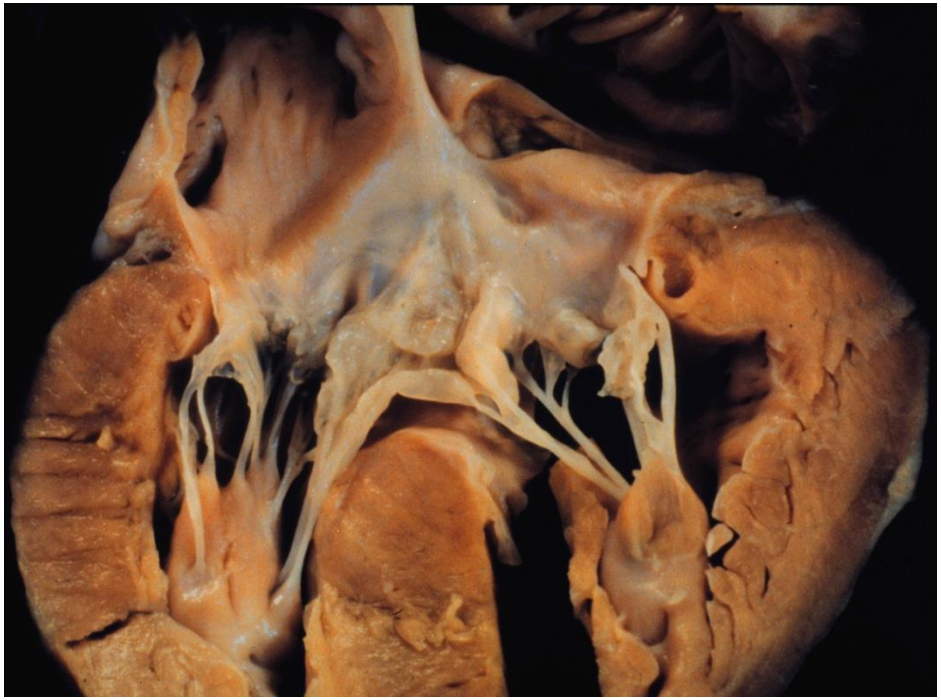
Ventricular Septal Defect



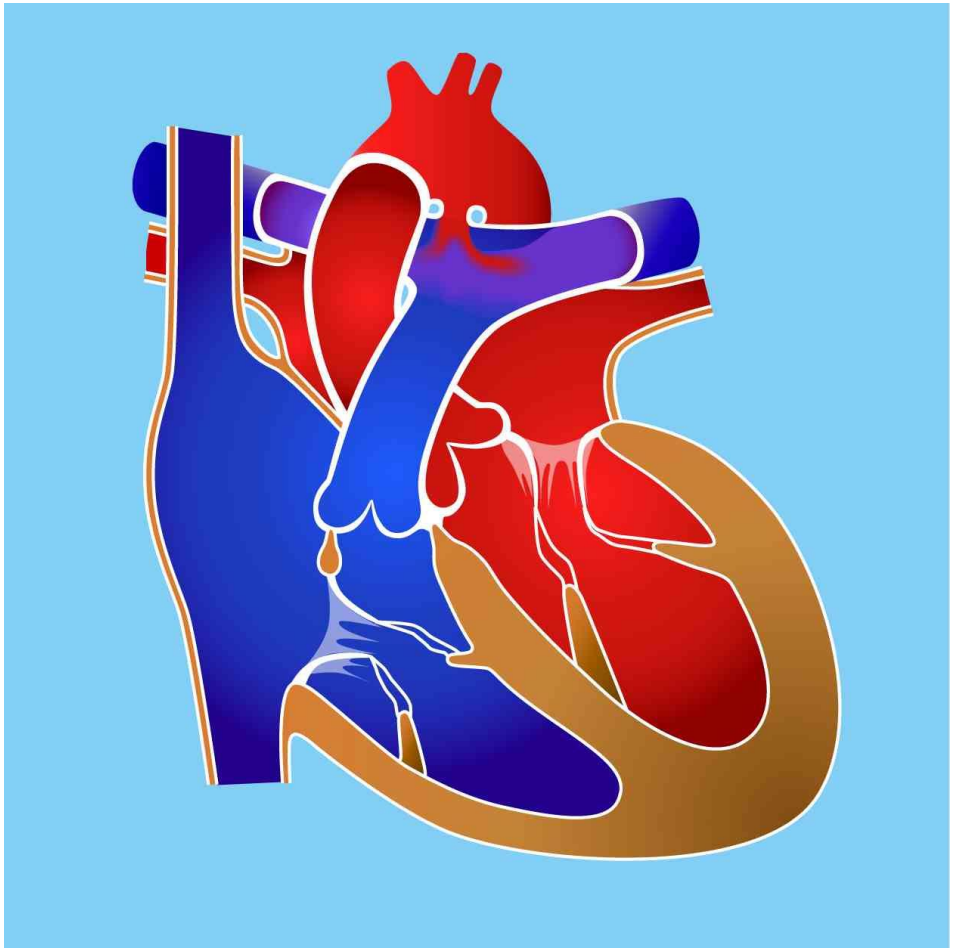


Atrioventricular Septal Defect





Persistent Arterial Duct



Left-to-Right Shunts (VSD, AVSD, PDA)

Increased pulmonary blood flow

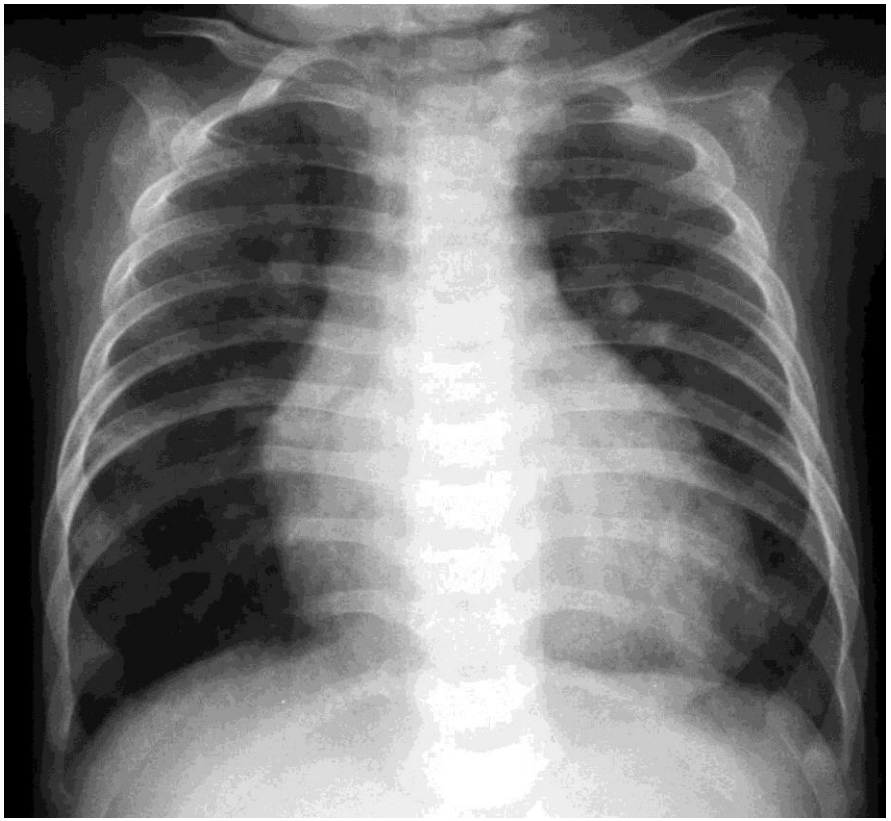
↓

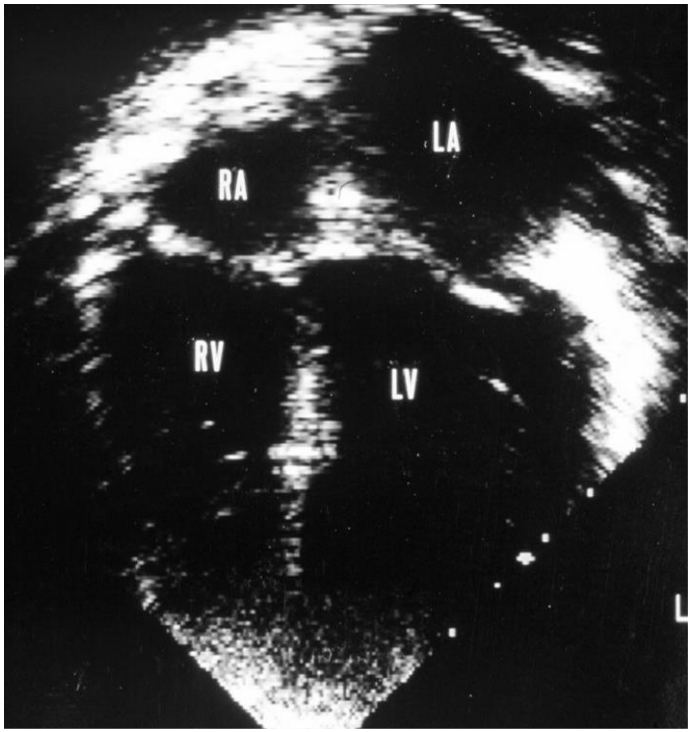
Increased pulmonary venous return

↓

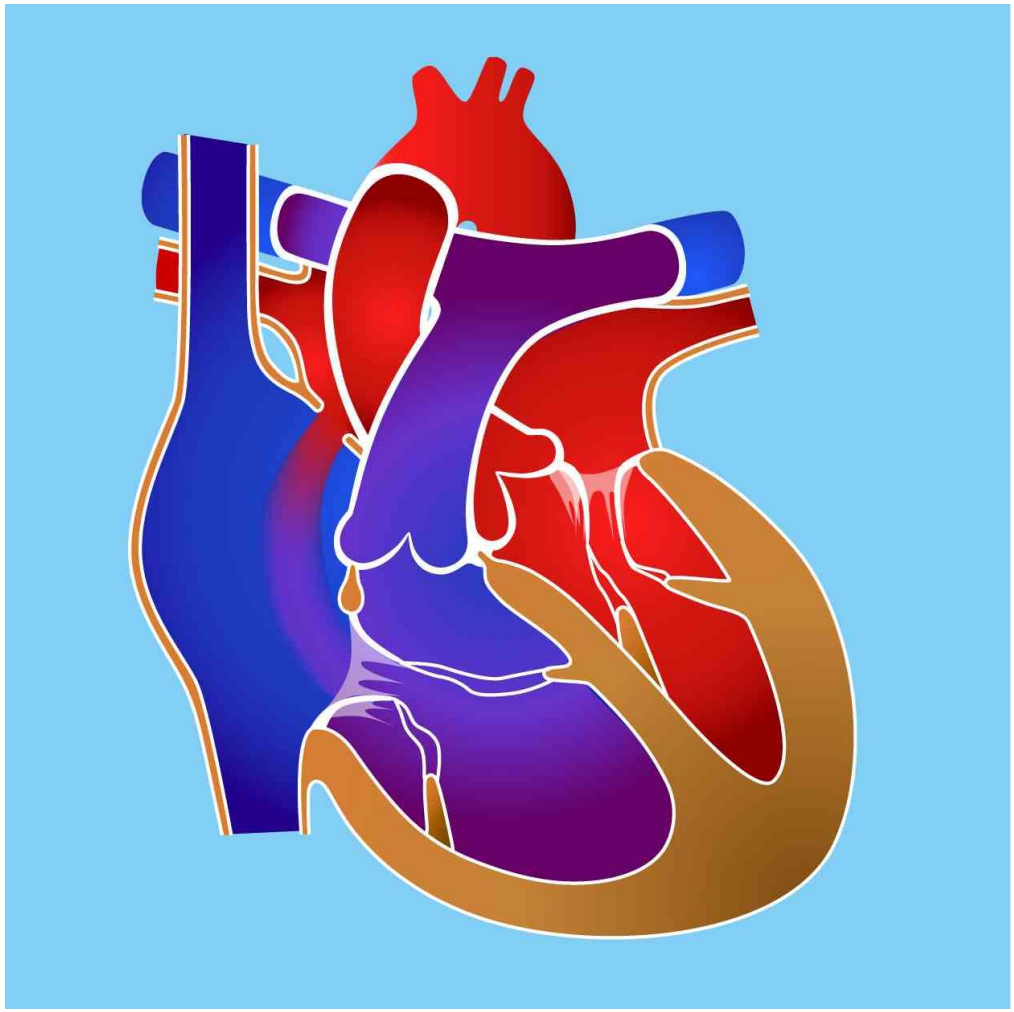
Volume overloading of left atrium and left ventricle (except atrial septal defect)

Left-to-Right Shunts





Atrial Septal Defect



Atrial Septal Defect

uncommon cause of heart failure in infancy and childhood

volume overloading of the right atrium and right ventricle

Left-to-Right Shunt (through ASD)

Shunting from LA to RA

↓

Increased flow through tricuspid valve during diastole

↓

Volume overloading of right ventricle

Heart Failure in Older Children

Myocardial disease

- myocarditis
- cardiomyopathy (primary, secondary)

unoperated structural heart defects

certain repaired or palliated congenital heart defects

Management of Paediatric Heart Failure

Identification of the cause and precipitating factors

Tackling of precipitating factors

General supportive management

Medical therapy of heart failure (diuretics, digoxin, ACEI, carvedilol)

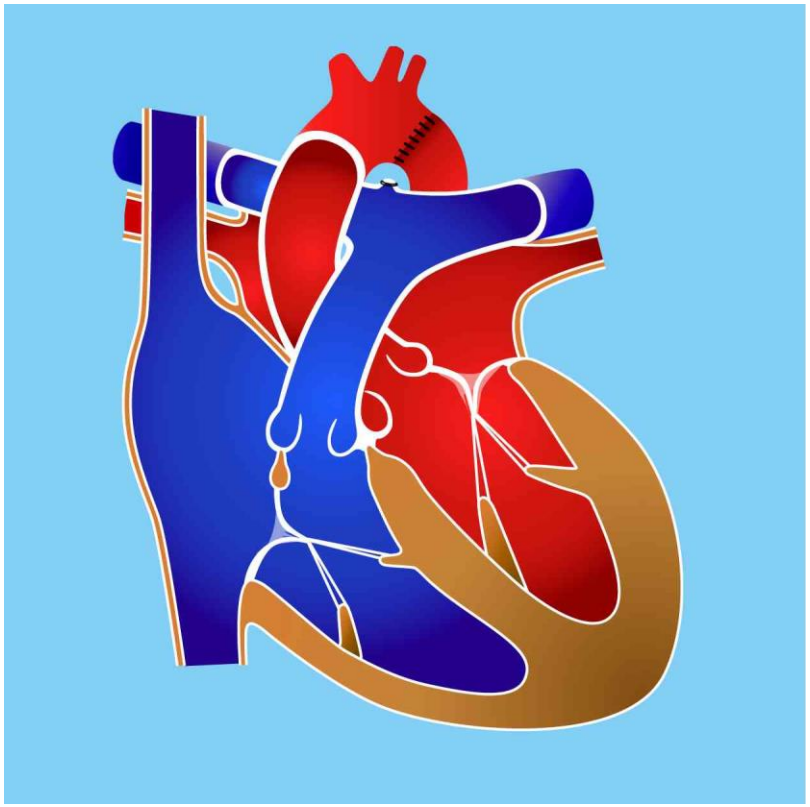
Treatment of underlying cause, if possible, by surgical or catheter intervention

Mechanical circulatory support and heart transplantation

Management of Severe Left Ventricular Outflow Obstruction

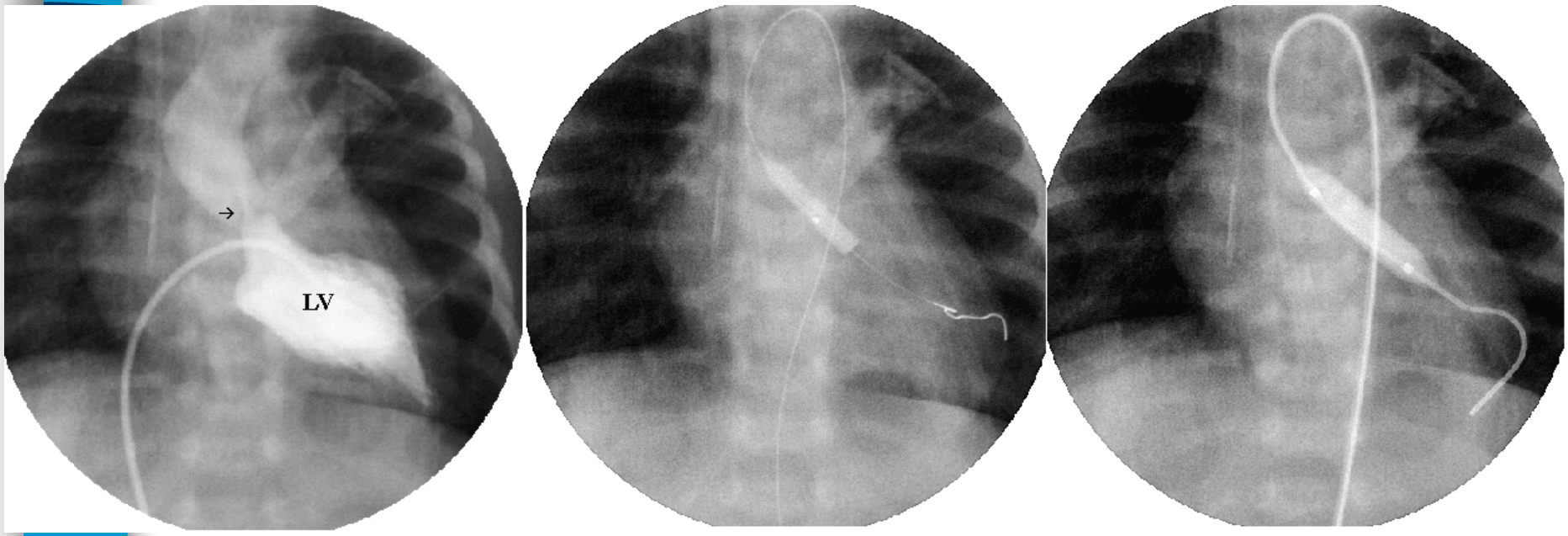
initial stabilization by PGE1/E2

corrective surgery or catheter intervention



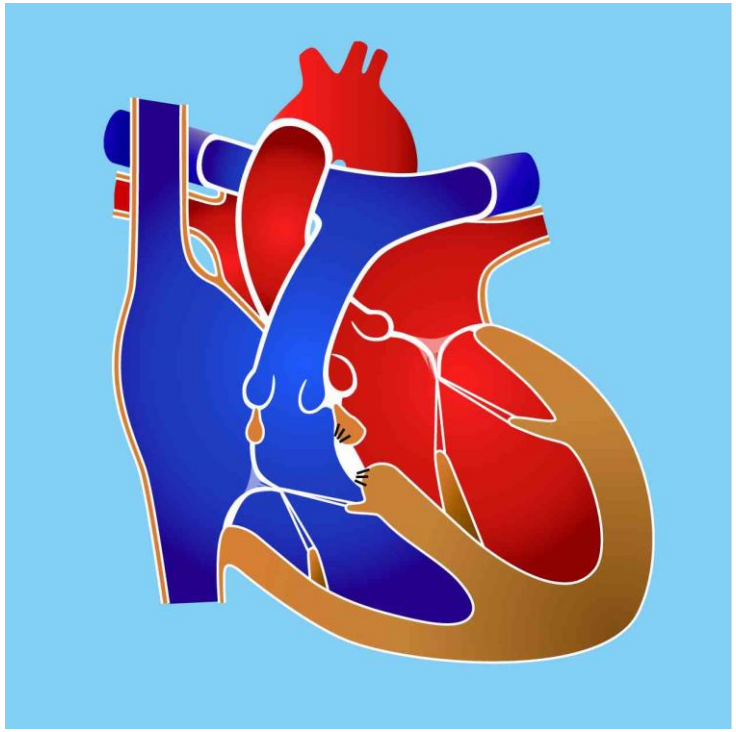
surgical repair of aortic coarctation / interruption

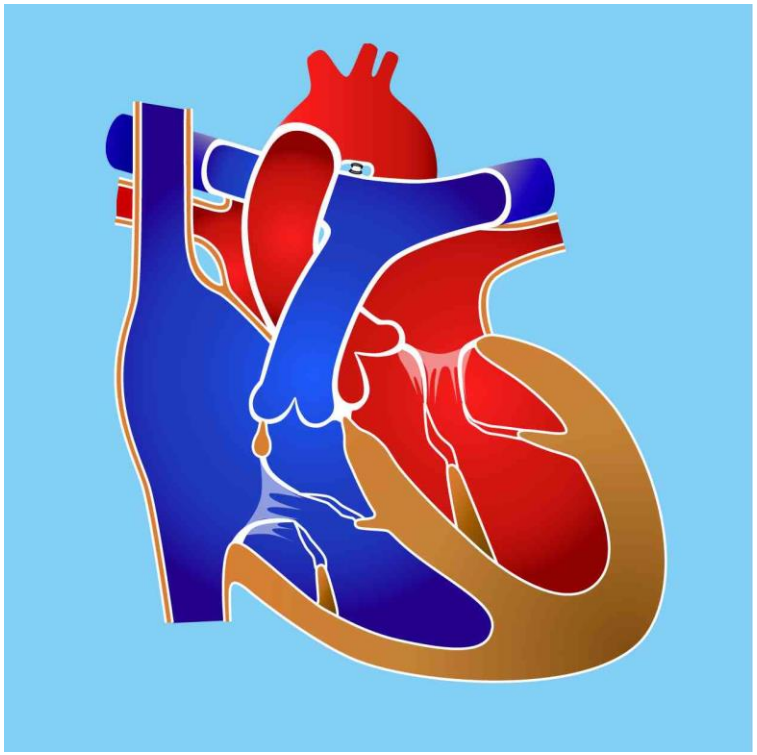
Management of Severe Left Ventricular Outflow Obstruction



Balloon dilation or surgical repair of critical aortic stenosis

Closure of Large Left-to-Right Shunts



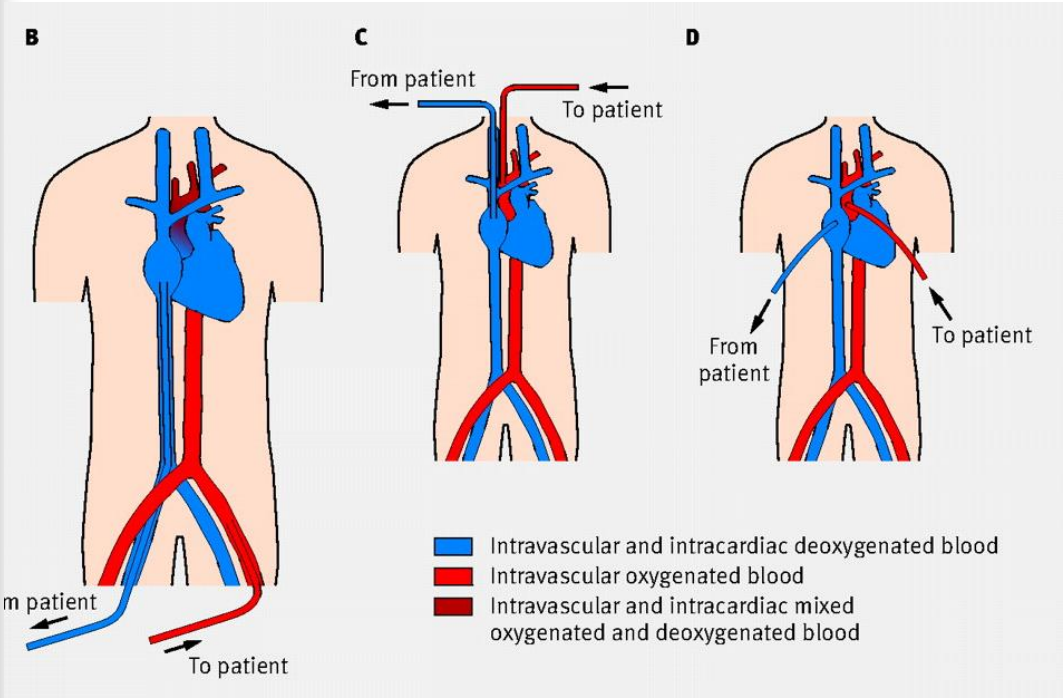


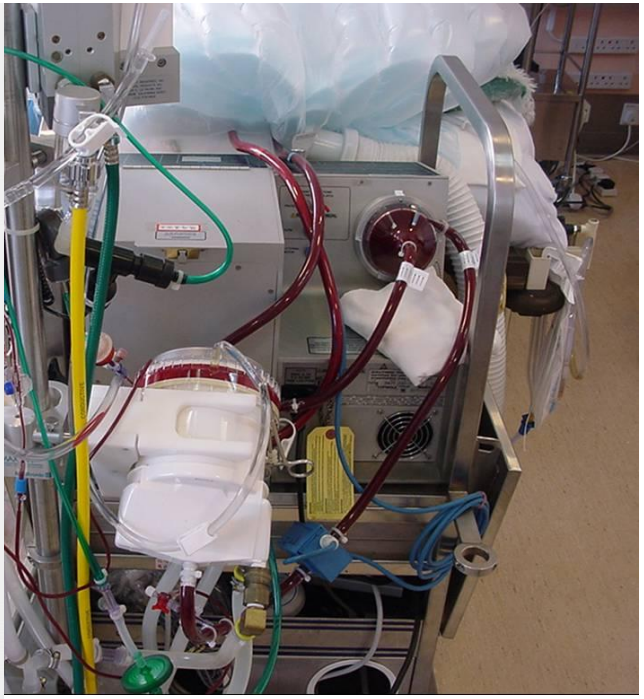
Surgical Closure of VSD and PDA

Closure of Large Left-to-Right Shunts

Device Closure of ASD and PDA

ExtraCorporeal Membrane Oxygenation

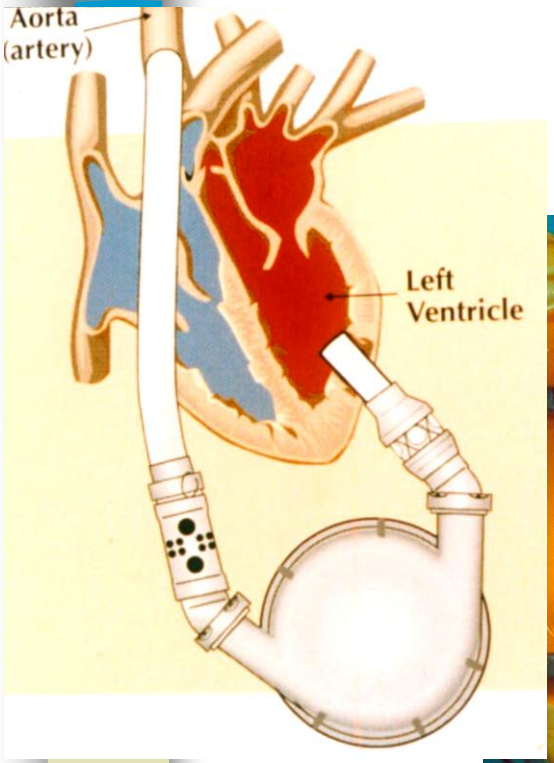


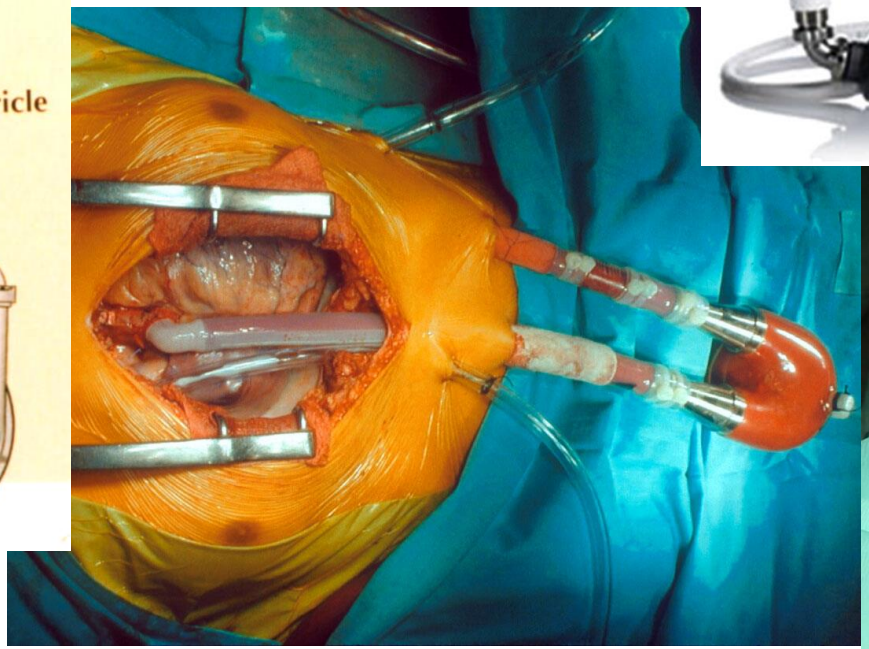


Gaffney et al. BMJ 2010

Ventricular Assist Device

Concept of Bridging





Overview

Concept of ‘heart failure’

Important categories of structural congenital heart lesions that cause heart failure in neonates and infants

‘Duct-dependent’ systemic circulation

Management of heart failure